

Primitive Roots of Unity

SEC 595: Secure Computing

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Objective

Find primitive roots of unity for all power-of-two subgroup sizes $N \in \{16, 8, 4, 2\}$ in the field $\mathbb{F}_{17}$.

1. Understand the Starting Field ($\mathbb{F}_{17}$)

- The prime is $p = 17$.
- The non-zero elements form the multiplicative group $\mathbb{F}_{17}^\times = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$.
- The size of this group is $p - 1 = 16$.

2. Find a Primitive Root for the Whole Field (ω_{16})

A primitive root (or generator) of $\mathbb{F}_{17}^\times$ is an element whose order is 16. That means its 16th power is 1, but no smaller power is 1.

Let's test $g = 3$ to see if it generates the entire group:

$$3^1 = 3$$

$$3^2 = 9$$

$$3^3 = 27 \equiv 10 \pmod{17}$$

$$3^4 = 30 \equiv 13 \pmod{17}$$

$$3^5 = 39 \equiv 5 \pmod{17}$$

$$3^6 = 15 \equiv -2 \pmod{17}$$

$$3^7 = -6 \equiv 11 \pmod{17}$$

$$3^8 = 33 \equiv 16 \equiv -1 \pmod{17}$$

Since $3^8 \equiv -1 \pmod{17}$, squaring it gives:

$$3^{16} \equiv (-1)^2 \equiv 1 \pmod{17}$$

Because 8 is the smallest power that gives -1 , 16 is the smallest power that gives 1. Therefore, the order of 3 is 16.

Result: 3 is a primitive 16th root of unity:

$$\omega_{16} = 3$$

3. Find the Primitive 8th Root (ω_8): We want an element ω_8 such that $(\omega_8)^8 \equiv 1 \pmod{17}$.

Since $\omega_{16} = 3$ has order 16, we can square it to get an element of order 8:

$$\omega_8 = (\omega_{16})^2 = 3^2 = 9$$

Verification:

- $9^1 = 9$
- $9^2 = 81 \equiv 13 \pmod{17}$
- $9^3 \equiv 13 \times 9 = 117 \equiv 15 \equiv -2 \pmod{17}$
- $9^4 = (9^2)^2 = 13^2 = 169 \equiv 16 \equiv -1 \pmod{17}$
- $9^8 = (-1)^2 \equiv 1 \pmod{17}$

Since $9^8 \equiv 1 \pmod{17}$ and no smaller power equals 1, 9 has order 8.

Result:

$$\omega_8 = 9$$

4. Find the Primitive 4th Root (ω_4): We want an element ω_4 such that $(\omega_4)^4 \equiv 1 \pmod{17}$.

Square ω_8 to halve its order from 8 to 4:

$$\omega_4 = (\omega_8)^2 = 9^2 = 81 \equiv 13 \pmod{17}$$

Verification:

- $13^1 = 13$
- $13^2 = 169 \equiv 16 \equiv -1 \pmod{17}$
- $13^3 \equiv -1 \times 13 = -13 \equiv 4 \pmod{17}$
- $13^4 = (13^2)^2 = (-1)^2 \equiv 1 \pmod{17}$

Since $13^4 \equiv 1 \pmod{17}$ and no smaller power equals 1, 13 has order 4.

Result:

$$\omega_4 = 13$$

5. Find the Primitive 2nd Root (ω_2): We want an element ω_2 such that $(\omega_2)^2 \equiv 1 \pmod{17}$.

Square ω_4 to halve its order from 4 to 2:

$$\omega_2 = (\omega_4)^2 = 13^2 = 169 \equiv 16 \equiv -1 \pmod{17}$$

Verification:

- $16^1 = 16 \equiv -1 \pmod{17}$
- $16^2 = (-1)^2 = 1 \pmod{17}$

Since $16^2 \equiv 1 \pmod{17}$ and $16^1 \not\equiv 1, 16$ (which is $-1 \pmod{17}$) has order 2.

Result:

$$\omega_2 = 16 \equiv -1$$

Summary Sequence of Example 5

Starting from the generator 3 of order 16, repeatedly squaring halves the order each time:

$$\begin{aligned}\omega_{16} &= 3 && (\text{Order } 16) \\ \omega_8 &= 3^2 = 9 && (\text{Order } 8) \\ \omega_4 &= 3^4 = 9^2 \equiv 13 && (\text{Order } 4) \\ \omega_2 &= 3^8 = 13^2 \equiv 16 \equiv -1 && (\text{Order } 2)\end{aligned}$$

A primitive root (or generator) of a finite field $\mathbb{F}_p$ is an element $g \in \mathbb{F}_p^\times$ whose powers generate every single non-zero element of the field.

In other words, if you take g and keep multiplying it by itself modulo p , you will visit all $p - 1$ non-zero elements before finally looping back to 1.

1. Formal Definition:

An element $g \in \mathbb{F}_p^\times$ is a primitive root modulo p if its multiplicative order is equal to $p - 1$:

$$\text{ord}(g) = p - 1$$

This means:

- (a) $g^{p-1} \equiv 1 \pmod{p}$ (guaranteed by Fermat's Little Theorem).
- (b) $g^k \not\equiv 1 \pmod{p}$ for any positive integer $k < p - 1$.

The set of all powers of g yields the entire multiplicative group:

$$\mathbb{F}_p^\times = \{g^0, g^1, g^2, \dots, g^{p-2}\}$$

- 2. Concrete Example ($\mathbb{F}_7$) For $p = 7$, the non-zero elements are $\mathbb{F}_7^\times = \{1, 2, 3, 4, 5, 6\}$, so $p - 1 = 6$.

Testing $g = 3$:

$$3^1 = 3$$

$$3^2 = 9 \equiv 2 \pmod{7}$$

$$3^3 = 3 \times 2 = 6$$

$$3^4 = 3 \times 6 = 18 \equiv 4 \pmod{7}$$

$$3^5 = 3 \times 4 = 12 \equiv 5 \pmod{7}$$

$$3^6 = 3 \times 5 = 15 \equiv 1 \pmod{7}$$

Look at the set of results: $\{3, 2, 6, 4, 5, 1\}$. Every single element of $\mathbb{F}_7^\times$ was generated, and the smallest power yielding 1 is 6. Therefore, 3 is a primitive root modulo 7.

Testing $g = 2$ (Non-Example):

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8 \equiv 1 \pmod{7}$$

The powers loop back to 1 after only 3 steps ($\text{ord}(2) = 3 \neq 6$). The powers of 2 only generate $\{2, 4, 1\}$, missing 3, 5, and 6. Therefore, 2 is NOT a primitive root modulo 7.

3. Connection to N -th Roots of Unity

In the context of the lecture handout (Sections 5 & 6):

- (a) A primitive root g of the field has order $p - 1$. It is a primitive $(p - 1)$ -th root of unity.
- (b) To find a primitive N -th root of unity ω_N (where $N \mid p - 1$), you simply take a primitive root g of the field and raise it to the power $\frac{p-1}{N}$:

$$\omega_N = g^{\frac{p-1}{N}} \pmod{p}$$

Example ($\mathbb{F}_{17}$ where $p - 1 = 16$): Since $g = 3$ is a primitive root of $\mathbb{F}_{17}$ ($\text{ord}(3) = 16$):

- Primitive 16th root: $\omega_{16} = 3^{\frac{16}{16}} = 3^1 = 3$
- Primitive 8th root: $\omega_8 = 3^{\frac{16}{8}} = 3^2 = 9$
- Primitive 4th root: $\omega_4 = 3^{\frac{16}{4}} = 3^4 = 81 \equiv 13 \pmod{17}$
- Primitive 2nd root: $\omega_2 = 3^{\frac{16}{2}} = 3^8 = 6561 \equiv 16 \equiv -1 \pmod{17}$